

STATEMENT OF WORK

1. SCOPE

1.1. Portsmouth Naval Shipyard (PNSY) is seeking to lease a temporary HVAC system, Exhaust Blowers, and Desiccant Climate Controlled Unit (DCCU) system for use at Naval Base Point Loma (NBPL) in San Diego, CA.

2. REFERENCES

2.1. **Issues of Publications** - The following documents in effect on the date of this requested proposal is a part of this specification to the extent applicable, or as specified herein. Within the specification, they shall be referred to by their basic designation only.

2.1.1. Government

2.1.1.1. US Federal Standard A-A-59326: Standard for cam action couplings.

2.1.2. Nongovernment

2.1.2.1. ARI Standard 410

2.1.2.2. Spark Resistant Construction (SRC: AMCA Standard 99-0401-86) Type C

2.1.3. **Order of precedence** - In the event of conflict between the text of this specification and that of the references cited above, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws or regulations unless a specific exemption has been obtained.

3. TECHNICAL REQUIREMENTS

3.1. **Component List**- The Contractor is required to bid on the equipment defined in 3.1

PIECE NO.	QTY.	NAME OF PART	TABLE NO.	Section
V-01	1	HVAC Unit: Dual Coil 20,000 CFM	1	4.3-4.5, 4.7, 4.9-4.15
V-02	4	Industrial Exhaust Blower	2	4.4, 4.9-4.15
V-03	2	DCCU: 15,000 CFM	3	4.3-4.5, 4.8-4.15
V-04	4	Electric Fan Coil Unit- Heater (150 KW)	4	4.4, 4.9-4.15
V-05	11	Remote Emergency Stop Switch	5	5.2.4.1.7
V-06	1	200/230 Ton Chiller	6	4.1-4.3, 4.5, 4.6, 4.9-4.15
V-07	1	600 GPM Chilled Water Pump	7	4.1-4.5, 4.9-4.15
V-08	1000 FT	6 IN Dia Chilled Water Hose	8	4.2, 4.3, 4.5
V-09	6	6 IN Dia. 90 Degree Elbows	8	4.2, 4.3, 4.5
V-10	400 FT	4 IN Dia Chilled Water Hose	8	4.2, 4.3, 4.5
V-11	12	4 IN Dia. 90 Degree Elbows	8	4.2, 4.3, 4.5
V-12	400 FT	3 IN Dia Chilled Water Hose	8	4.2, 4.3, 4.5
V-13	12	3 IN Dia. 90 Degree Elbows	8	4.2, 4.3, 4.5
V-14	1	6 IN Dia Globe Valve	8	4.2, 4.3, 4.5
V-15	4	Chilled Water Manifold	8	4.2, 4.3, 4.5

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Table 1: Dual Coil HVAC Unit	
Maximum Dimensions	21 ft long, 8.5 ft wide, 18,000 lbs
Emergency Stop	Must be pre-wired for, compatible with, and supplied with, Emergency Remote Stop (E-Stop) (See Table 5 for E-Stop Requirement)
Fabrication	All components must be fully integrated within single housing equipped with appropriately rated overhead lifting points.
Power Supply	480 VAC Nominal / 3 Phase / 400FLA max.
Electrical Connections	Camlok Style (J Series locking style i.e. 1016 series)
Process Air Minimums	Four outlets. (All outlets must be 20" diameter, each, capable of connecting 20" flex duct).
	20,000 CFM (5,000 per outlet) at 7 inches External Static Pressure
	TEFC Premium Motors, VFD fan controller capable of $\leq 50\%$ of rated airflow
	Tightly sealed non-detachable process air inlet/outlet. Detachable covers are permitted.
Process Air Filtration	Moisture Separator/Demister type filtration for purging 98% of the moisture. Paper or fibrous filtration media is NOT ACCEPTABLE.
	Metal process air inlet filters
Electric Heater Minimums	240KW electric heating element, thermostatically controlled to discharge air temperature set point.
Cooling Coil Ton (min)	120 Tons with 42F entering chill water and 10F ΔT thermostatically controlled.
Cooling Coil Connection	4" non-ferrous, non-galvanized, metal cam and groove quick-disconnect style per reference 2.1.1.1, with manual air bleed valves

Table 2: Industrial Exhaust Blowers	
Physical Attributes	8'-6"L & W x 9'H
Minimum Capacity	10,000 CFM at 14 inches External Static Pressure
Electrical	480 VAC Nominal/ 3 Phase, with appropriately sized disconnect switch. 100FLA max.
Motor	NEMA Design B, TEFC premium, industrial, continuous-duty, ball bearing, foot mounted. Bearings shall be rated L-10 life (40,000 hrs). Equipped with VFD.
Safety	Must be pre-wired for, compatible with, and supplied with, Emergency Remote Stop (E-Stop) See Table 5
	Must include guards for drive assembly per OSHA regulations and suction/discharge outlet fan guards
	Vibration Sensor with programmable electronic automatic shutdown, for protection of equipment. Sensor shall have two setpoints: lower setpoint illuminates a "caution" light. Upper setpoint secures blower automatically and illuminates a "warning" light.
Construction	AMCA Standard 99-0401-86 Type C, Centrifugal, Up-Blast. Fan Wheel must be constructed with flat blades. Airfoil and backward incline blades are not acceptable. Blades shall be continuously welded to reduce stress.
Connections	Adaptable to two 14" inlets or six 8" inlets, capable of connecting 14" or 8" flexible duct. Contractor is responsible for providing transition inlet assembly.
Cleanout	Gasketed door, either hinged or bolted, to provide maintenance and cleanout access. Drain pipe connected to lowest point on fan housing, due to up-blast configuration.
Discharge	Either Noise Attenuator, or discharge duct that doubles its cross section over four feet length

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Table 3: Dehumidification Unit	
Maximum Dimensions	21 ft long, 8.5 ft wide, 20,000 lbs.
Emergency Stop	Must be pre-wired for, compatible with, and supplied with, Emergency Remote Stop (E-Stop) (See Table 5 for E-Stop Requirement).
Fabrication	All components must be fully integrated within single housing equipped with appropriately rated overhead lifting points.
Power Supply	480 VAC Nominal / 3 Phase / 400FLA per circuit, two circuits max.
Electrical Connections	Camlok Style (J Series locking style i.e. 1016 series).
Process Air Minimums	Two Outlets. (All outlets must be 20" diameter, each, capable of connecting 20" flex duct).
	5,000 CFM per outlet at 10 inches External Static Pressure.
	TEFC Premium Motors, VFD fan controller capable of $\leq 30\%$ of rated airflow.
	Tightly sealed non-detachable process air inlet/outlet. Detachable covers are permitted.
Reactivation Air (Min)	4200 CFM.
Process Air Filtration	30% efficiency or greater. Clean (reusable) or new (disposable) at time of delivery.
Electric Heater Minimums	Reactivation Heater: 232 KW Electric Heating Element (no backup heat source).
	Post-Heater: 312 KW Electric heating element, thermostatically controlled to discharge air temperature set point.
Dehumidification	≥ 220 Lbs./Hr. per Outlet (≥ 440 lbs/hr with 2 outlets, ≥ 660 lbs/hr with 3 outlets).
Cooling Coil Connection	Pre-Cooling: 4" non-ferrous, non-galvanized, metal cam and groove quick-disconnect style per reference 2.1.1.1, with manual air bleed valves.
	Post-Cooling: 3" non-ferrous, non-galvanized, metal cam and groove quick-disconnect style per reference 2.1.1.1, with manual air bleed valves.
Cooling Capacity	Pre-Cooling: 50.8 Tons @170 GPM.
	Post-Cooling: 32.1 Tons @ 93 GPM, thermostatically controlled to discharge air temperature set point.
	Overall: 100 Tons, nominal, with 45°F entering chill water and 10°F ΔT .

Note: Moisture Removal Rate based on [80°F EDB / 71°F EWB & 75°F LDB / 17% RH]

Table 4: 150KW Electric Heaters	
Maximum Dimensions	4'W x 8' L x 5' H. 2500lbs
Minimum Airflow	7000 CFM at 5.6 inches Total Static Pressure
Minimum Heat	150KW electric heat with 240F air temperature rise
Motor	TEFC Premium Motors with VFD Controller
Electrical	480 VAC Nominal / 3 Phase / 200FLA max.
Electrical Connections	Camlok Style (J Series locking style i.e. 1016 series)
Controls	Thermostatically controlled discharge air temperature set point
Safety	Must be pre-wired for, compatible with, and supplied with, Emergency Remote Stop (E-Stop) See Table 5
	Safety controls integrated into controller
Connection	One 20 inch diameter outlet capable of connecting to 20 inch flexible ducting.
Construction	Must include fork pocket slots and overhead rigging connections (padeyes)

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Table 5: Emergency Remote Stop (E-Stop)

Cable Type	Type SO or equivalent (Heavy duty, flexible, portable, durable)
Cable Length	1500 ft per E-Stop to accommodate tortured routing paths
Accessories	Plexiglas cover for all E-Stop buttons to prevent inadvertent activation

***NOTE:** Contractor is responsible for installation and routing of all E-Stop components.

Table 6: 200-230 Ton Air Cooled Chiller Unit

Maximum Dimensions	27.5 ft long, 8.5 ft wide, 22,000 lbs
Safety	Must have a low-flow switch (<i>not</i> a pressure differential switch).
Fabrication	All components (with sole exception of the pump) must be fully integrated within single housing equipped with appropriately rated overhead lifting points.
Electrical	480 VAC Nominal / 3 Phase / 400FLA max per circuit. Integrated chiller pump MUST be powered separately.
Electrical Connections	Camlok Style (J Series locking style i.e. 1016 series)
Cooling Capacity	≥210 Ton heat rejection @ 95°F Ambient conditions
Leaving Water Temp	42±2°F
Chill Water Outlet Range	38°F-60°F
Condenser Type	Air Cooled, forced air
Chiller Components	Minimum of 2 separate compressor and refrigeration circuits
Chill Water Connection	6 inch, non-ferrous, non-galvanized, metal cam and groove quick disconnect fittings per reference 2.1.1.1.
Chiller Interface	Must have a display which allows end-user to jog through current and historical data.
Data Requirements	System Performance Data
	System Parameters Adjustment
	System Errors (Recorded and logged during equipment failure)
Flow Meter (Calibrated)	Electronic 100-800 GPM @ 1 GPM resolution Electronic in-line, downstream of pump designed for system working pressure.

Table 7: Chiller Pump

Configuration	Can be integral to the Chiller, or standalone, with same connection requirements as Chiller.
Performance	600 GPM at 100 ft head (H ₂ O) compatible with chiller unit, if standalone.
Motor	Low speed, 1750 RPM
Electrical	480 VAC Nominal/ 3 Phase, capable of being powered independently of Chiller with appropriately sized disconnect switch.
Safety	Differential Pressure gage rated from 0-150 PSI, or Max discharge pressure +20 PSI, whichever is greater.

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Table 8: Hoses & Fittings	
Hose Type	6", 4", and 3" non-collapsible suction/discharge hose.
Hose Lengths (max)	25 ft
Connection Types	Male & Female Cam and Groove quick disconnect fittings, per reference 2.1.1.1 supplied with new gaskets for all hoses and manifolds.
Pressure Rating	150 PSI
Manifold	Cross-style, with two 6" connections (male & female) and two 4" connections (female).
	Sufficient adapters to convert half of 4" female to 4" male.
	All connections must have hydrostatic caps/plugs capable of sealing against system pressure.
	All 4" legs shall have isolation valves.
	Manifolds shall be designed such that they stack together to minimize laydown space.
Transition fittings	Adapter fittings to attach 4" hose to 3" equipment connections required if any equipment is provided with 3" connections.
Test Requirements	All hoses and fittings must have been previously hydrostatically tested to 1.5 times the pressure rating within 1 year of issuance.
Spare Soft goods	All hoses and fittings must be supplied with replacement gaskets for each fitting/hose size.

4. **NOTES:**

4.1. **Chiller Capacity & Utilities:**

- 4.1.1. **Cooling:** The primary utility for the Chiller shall be all electric and air-cooled. The minimum nominal chilled water capacity shall be as indicated with the actual capacity, based on an ambient temperature of 95°F dry bulb with an Entering Water Temperature (EWT) of 52°F and a Leaving Water Temperature (LWT) of 42°F, to be not less than 5% of the desired nominal capacity. The unit capacity shall not be greater than one-third above (+1/3) the minimum specified capacity and shall be able to run continuously at one-quarter of that unit's stated capacity.
- 4.1.2. **Automatic Operation, Regulation, & Controls:** The Chilled Water Unit or Chiller Module shall be equipped with an automatic variable capacity control requiring only initial temperature adjustment of from 38 to 60°F by the operator. Refrigeration circuits shall maintain the specified temperature condition while protecting against coil freeze-up. The unit / modules shall be completely wired for safety, capacity, and operation such as a power/switch system with circuit breakers, manual/automatic controller, motor overload protection, high/low pressure cut-outs, energy control system, and other essential components. The unit / modules shall be fully controllable under a wide range of loads even though the daily and seasonal ambient temperatures vary greatly.
- 4.1.3. **Design & Component Integration:** The unit / modules shall be designed for continuous operation at minimal loading (i.e. one-quarter of full capacity for refrigeration & one-third of full capacity for pumping) despite extreme ambient conditions (e.g. -10 to +100°F and 0 to 100% RH).
- 4.1.4. **Cooling:** The condenser(s) shall be air-cooled via fan(s). The chiller shall have a sufficient water storage capacity to run isolated from a make-up water connection.
- 4.1.5. **Energy Efficiency:** For energy efficient operation in a fully loaded condition, the Full Load Amperage (FLA) rating for the combined Chilled Water Unit or the FLA summation for a separate Chiller Module plus one Pump Module shall be two and a half amperes or less per ton of actual chilled water capacity.
- 4.2. **Coolant Free:** Besides manifolds and hose assemblies that are normally flushed and tested, all machinery conveying chilled water such as: the Chiller, Pump and machines with a chilled water coil; shall arrive with no coolant of any type in its piping (i.e. No Water & No Anti-Freeze).
- 4.3. **Chilled Water Hose Assembly / Manifold Construction / Cooling Coil Hydrostatic Testing:** For standard Chilled Water Hose assemblies of six inch nominal diameter or less, the exterior cover shall be a thick rubber (Buna-N) type material designed as a CHW insulator. All Chilled Water connectors / fittings must be quick disconnect style fittings meeting the requirements defined in reference 2.1.1.1. Chilled Water Hose assemblies shall be non-collapsible and

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the camlock fittings shall be attached to the hose material via the more secure, newer type, crimp style hose fitting via a single specified diameter, elongated sleeve length and not from the less secure, older type, band clamp style hose fitting via multiple universal diameter, shorter ferrules. Gaskets for couplings shall be pre-installed inside the female coupling(s) of hose assemblies and manifolds. The gasket shall be inspected for presence, tight fit, and lack of deterioration. Gaskets shall not protrude into the waterway.. Hose/manifold/valve assemblies and all cooling coils shall be hydrostatically tested with new hose gaskets for Strength and Tightness and/or Valve Seat Leakage by an independent (of the Contractor) and certified Contractor to a minimum of 150 psig for five minutes and visually inspected and/or internal valve leakage measured while pressurized. Hydrostatic test date shall be within one calendar year of scheduled Arrival Date of the equipment. Hoses/manifolds/valve assemblies shall have a banded identification (ID) tag or label attached to the assembly displaying the hydrostatic test results. The Hydrostatic Test Certificates issued by the certified testing facility shall provide a copy of the certificates to PNSY Temporary Services Design Engineer and the PNSY Contracting Office.

- 4.4. **Phase Rotation:** To prevent or identify an out-of-phase electrical power situation, three-phase machines shall include either: **(a)** a reverse phase relay and/or meter that identifies an out-of-phase condition and prevents operation especially applied to complex machinery in which such a condition could cause damage or **(b)** an arrow prominently displayed at an appropriate location on the machine's housing to indicate the proper direction of shaft rotation with the ability to observe a portion of the shaft after a momentary application of power (a.k.a. jog or bump). A reverse phase relay is not required nor desired for VFD motors, however, the VFD shall have a jog button so as to verify correct rotation.
- 4.5. **Supplemental Equipment:** For the protection and cleanliness of the machinery components, all chilled water connection points shall be shipped with plug/cap fittings.
- 4.6. **Hydronic Flow Meter:** Attached to the Chiller and its associated pump shall be a continuous reading liquid flow meter, preferably, in-line with flow. The type of meter may be Electronic Metering. The physical positioning of the instrument shall be downstream of the Chiller immediately after its discharge. The flow meter assembly shall be rated for 150 psig;. The instrument shall be calibrated by a licensed Metrology Laboratory within five years of its scheduled delivery with certified calibration test results both accompanying the flow meter assembly and reported via E-mail to the Contracting Office Representative (COR).
- 4.7. **HVAC Capacity & Utilities**
 - 4.7.1. **Control:** The unit shall include a permanently mounted control board accessible without opening any panels with temperature thermometer(s) and pressure gauge(s) as necessary for monitoring and control. The unit shall be fully controllable under a wide range of loads even though the diurnal and seasonal ambient temperatures vary greatly.
- 4.8. **DCCU Capacity & Utilities:**
 - 4.8.1. **Moisture Removal:** As a stand-alone unit, the DCCU shall be capable of delivering processed air at 0°F dew point or lower when the ambient conditions approach saturation at a temperature of 75°F. The minimum rate for moisture removal at the specified full rate of airflow is based on ambient conditions as synopsized below of either:
{75 degrees Fahrenheit at 50% Relative Humidity (75°F @ 50% RH)} or
(80 degrees Fahrenheit at 60% Relative Humidity (80°F @ 60% RH)) or
[90 degrees Fahrenheit at 120 grains per pound of dry air Absolute Humidity (90°F @ 120 gr / lb. AH)]
 - 4.8.2. **Control:** The unit shall include a permanently mounted control board accessible without opening any panels with temperature thermometer(s) and pressure gauge(s) as necessary for monitoring and control. The unit shall be completely wired for safety, capacity, and operation such as a power / switch system with circuit breakers, manual / automatic controller, motor overload protection, high / low pressure cut-outs, energy control system, and other essential components. The unit shall be fully controllable under a wide range of loads even though the diurnal and seasonal ambient temperatures varies greatly.
 - 4.8.2.1. **Moisture Removal:** The reactivation wheel shall be capable of being manually shut-off and still be able to operate the process air blower.

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- 4.8.3. **Design & Component Integration:** The unit shall be designed for continuous operation despite extreme ambient conditions (e.g. -10 to +100°F & 0 to 100% RH). The desiccant dryer shall be an adsorption type wheel with a single desiccant rotary structure. Desiccant wheel matériel can be either the former titanium silica-gel or the more recent zeolite or even a more advanced superadsorbent Polymer, but not the older more problematic lithium chloride.
- 4.8.4. **Filtration:** The air to be processed shall be pre-filtered and its filtration efficiency shall be 30% or greater. The unit shall be delivered with all clean (reusable) and/or all new (replaceable) filters. The filters shall be of the correct dimension and in sufficient quantity to completely fill the design space of the filter rack(s) with no discernible gaps. Prior to start-up, the Contractor shall demonstrate that all reusable filters are clean & replaceable filters are new and are properly installed. For dehumidifiers, the air to be filtered is at or near the intake(s) to the Process Air Inlet(s).
- 4.8.5. **Supplemental Equipment:** For the protection and cleanliness of the machinery components, all air and water/steam connection points shall be shipped with sheet metal covers (air) and plug/cap fittings (water/steam).
- 4.9. **Construction:** Except as noted for a particular machine type, units shall be a single integrated framed structure manufactured as a complete entity to include all devices, accessories, and fittings necessary to perform the function for which it was designed. Each unit shall be capable of continuous, year-round, outdoor, unmanned operation even at minimal loading while exposed from inclement to extremely harsh weather (e.g. deep snow, strong & gusty winds, and/or heavy rain) and varied ambient conditions (-10 to +110°F & 0 to 100% RH). Machines shall meet State and federal requirements for its type such as the National Electrical Code (NEC) for electrical devices / components and the American Society of Mechanical Engineers (ASME) for pressurized vessels / components. Machines shall be modular in design so as to act as a stand-alone unit; compact so as to fit in the tight confines of a drydock; portable so as to be transported by forklift and then lifted by crane; and ruggedly constructed so as to endure a marine industrial environment. Machines shall be newly manufactured / completely refurbished and factory tested within ten (10) years.
- 4.10. **Electrical:** All equipment shall be Nationally Recognized Testing Laboratory (NRTL) listed and labeled. Electrical components shall conform to a NEMA standard (NEMA 3 or 4) that allows for unsheltered outdoor use of the unit. Each machine shall require only one type / source of electrical power (480 Nominal VAC / three-phase / 60 hertz). The type of electrical power required (voltage & amperage) must be clearly indicated on the outside of the equipment, readily detected, and collocated with all other electrical information. The Contractor shall provide the PNSY Contracting Office with the following details: (i) electrical power type (amperage & voltage); (ii) confirming that the type of electrical connectors for three phase shall be camlok; (iii) breaker type/size or recommended capacity for the electrical source; and (iv) whether the machine is furnished with over-current protection.
- 4.11. **Lifting & Handling:**
- 4.11.1. **Lifting & Handling Exception:** Although considered to be machinery, individual pump units, being relatively lightweight and compact, are not required to be enclosed in a caged frame. However, the remaining Lifting and Handling requirements pertain in that the unit will be handled by forklift.
- 4.12. **Noise Requirement:** When operating at full speed, noise levels must not be over 85 dB[A] at less than three feet from the unit or frame unless a quieter noise level is specified. A hearing protection safety placard shall be posted on each machine generating more than 85 dB[A] at the unit's edge or frame.
- 4.13. **Machinery / Equipment Tagging and Markings:** Fluid piping inlet(s) / outlet(s) must be clearly marked. Machinery and itemized equipment require tagging to show Project Number / Technical Specification Identification (ID) Number
- 4.14. **Contractor Identification of Itemized and Ancillary Equipment:** The Contractor shall uniquely tag and permanently mark all of its ancillary and itemized equipment / hardware (e.g. manifolds, hose assemblies, pipe fittings, duct adapters, ducting, etc.) so as to differentiate from that owned by either the Shipyard or other Contractors.

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- 4.15. **Dead Front Control Panel or External Controller:** Access to basic machinery controls, such as a Dead Front Control Panel or External Controller, shall be such that there is non-exposure to energized *"live"* electrical circuits or other components regardless of voltage, to include control circuitry, by *"a person on the operating side of the equipment"*.

5. RESPONSIBILITIES

5.1. Government

- 5.1.1. Equipment Technical Evaluation: Temporary Services Engineering
- 5.1.2. Receipt Inspection: Contractor Officer Representative (COR)
- 5.1.3. Coordination of equipment delivery & returns: COR
- 5.1.4. Schedule equipment service, repair, and training: COR

5.2. Contractor

- 5.2.1. Use Table E1 of Enclosure 1 to verify conformity with the Technical Requirements.
- 5.2.2. The equipment will be operating around-the-clock under a rigorous deployment performance schedule. Within twenty-four (24) hours of initial notification of the need for repair, the Contractor shall have a knowledgeable and equipped technician on the job-site fully prepared to repair the rental equipment. The defective unit(s) shall be repaired within forty-eight (48) hours of initial notification or, if the original defective unit(s) cannot be made operational, replaced within seventy-two (72) hours of initial notification. If the equipment failure is not the fault of the Government, the Contractor shall bear all costs associated with its repair and/or replacement.
- 5.2.3. The contractor shall provide technical support to PNSY project personnel as required.
- 5.2.4. Equipment Start-up/ Commissioning Services
 - 5.2.4.1. Contractor must provide start-up and commissioning services to prepare the equipment to be placed into service. This includes but is not limited to the following:
 - 5.2.4.1.1. Pressurizing the system.
 - 5.2.4.1.2. Performing leak check of all contractor made connections.
 - 5.2.4.1.3. Adjusting setpoint(s) to meet the requirements of this Statement of Work.
 - 5.2.4.1.4. Verifying satisfactory operation of all drains installed on equipment.
 - 5.2.4.1.5. Troubleshooting/adjudicating any alarm conditions present on start-up (See 5.3).
 - 5.2.4.1.6. Training Government operators in equipment function, control, and alarm(s)/indications.
 - 5.2.4.1.7. Installation and routing of all E-Stop components.

5.3. Maintenance

- 5.3.1. Except for daily surveillance checks, the Contractor is responsible for all equipment preventative maintenance and repairs for the duration of the lease. Coordinate maintenance with Contracting Office Representative (COR) or Technical Point of Contact (TPOC).
- 5.3.2. The contractor shall provide Preventative Maintenance (PM) schedules which are equipment specific. The PM shall meet all requirements determined by the Original Equipment Manufacturer (OEM). The contractor shall provide the PM documentation and schedules to the COR/TPOC for review and approval prior to being considered for a successful Bid. The routine PM plan for each piece of equipment shall include a comprehensive list of the components to be serviced as well as identifying the service frequency. (e.g. all consumables to be replaced, bearing grease points, belt replacement / tensioning, filter changes, refrigerant charge, oil changes, other lubrication, coolant refills, liquid top-offs, etc.) All PMs conducted shall include a technical service report identifying component, maintenance conducted, date, and next scheduled PM date. Copies of all reports shall be delivered to the project COR and the TPOC.

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5.3.3. In addition to providing routine preventative maintenance, the contractor shall provide a service technician on a weekly basis for the duration of the lease to perform weekly equipment inspections. Weekly equipment inspections shall consist of checking all functional operating parameters of each individual piece of equipment. This shall include but is not limited to: verifying correct equipment performance and operation, checking for diagnostic or trouble codes, measuring discharge air temps, verify amperage of all electric heating coils, verification of chilled water coils performance, verify pump operation, motor speed, and differential pressure, verify HVAC process air fan speeds, verify HVAC intake filter condition, Chiller performance operation, etc. Weekly equipment inspections shall include a technical field report identifying the status of each piece of equipment and shall include all data / measurement points verified. All reports are delivered to the project COR.

5.3.4. The contractor shall provide a knowledgeable service technician to cover the repair of all equipment deficiencies and malfunctions. The response time for the repair or replacement of deficient equipment shall meet the requirements defined in section 5.2.1.

5.4. Training

5.4.1. The Contractor is responsible for and shall provide training sessions to Shipyard Personal on the operation & setup of all equipment. Training shall be available throughout the duration of the lease to accommodate equipment swap out and seasonal equipment operation.

5.5. **Deliverables** – In addition to all required equipment, technical documentation shall also be provided electronically to the project COR and Engineering. Minimally, the documentation shall consist of:

5.5.1. Lifting & Handling the equipment with its padeye(s) Certification.

5.5.2. Points of contact with telephone numbers for requesting emergency repair regardless of the time or type of day.

5.5.3. Technical Instructions (TIs) such as an Operations & Maintenance Manual, Controls Manual, and/or Mechanical/Electrical Manual with Schematics.

5.5.4. Operating Instructions, if not in the TI.

5.5.5. Preventative Maintenance Schedules. (See section 5.3.2)

5.5.6. Equipment Data Sheet (EDS).

5.5.7. Hydrostatic Pressure Test Reports.

5.5.8. Flow meter calibration certificate.

6. DELIVERY / ACCEPTANCE / RETURN

6.1. **Delivery** - The machinery shall be in a fully charged, fully assembled, ready-to-use, and first-class operating condition. Machines shall be functionally tested prior to their delivery, and all leased products, including machine / equipment / ancillary component accompanying machine, must fully function as intended. The machinery with its support equipment shall be delivered to the site on or prior to the request date. It is incumbent upon, and the responsibility of, the Contractor to either confirm or coordinate the date and time of delivery prior to each shipment via the Point-of-Contact (POC) as designated on the Contract and/or the Contracting Officer's Representative (COR). The Government can provide forklift services for removing machinery from Contractor's transport during work hours. Documentation with Shipyard copies shall accompany each delivery.

6.2. **Acceptance**- Upon satisfactory completion of inspection and test of the equipment, the authorized Government Representative (Portsmouth Naval Shipyard, Equipment Manager) will sign and forward the acceptance document(s) to the POC as designated on the Contract and/or COR for completion. Unloading of equipment from Contractor's transport does not signify acceptance.

6.3. **Start Up**- Upon delivery, the Contractor shall: (i) provide start up and commissioning of the equipment to verify its operability per SOW section 5.2.4; (ii) provide all deliverables as required per SOW section 5.5 and (iii) train designated Shipyard personnel on how to properly run the equipment to include its operation, control, daily checks, & service contacts per SOW sections 5.2.4.1.6 and 5.4.

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6.4. Security - To be allowed onto PNSY:

- 6.4.1. All personnel assigned by the contractor to perform work or site visits must be US citizens. The contractor company manager will be required to submit a visit request listing information of personnel needing access aboard Portsmouth Naval Shipyard. Visit request must include the full name of the persons nationality, birth date and Social Security Number. Visit request may be in the form of a letter on company stationery with signature from company security manager or authorized company representative and faxed to the Pass and ID office at (207) 438-2669 or emailed to PNSDBIDS@US.NAVY.MIL. Company visit requests shall be submitted annually and as often as necessary to update employees of the company who need continued access to the Shipyard. Criminal history and/or falsification of information provided by an employee of the company may be subject to denial for unescorted access up to and including general access to the Shipyard. For contractors needing short term access to the shipyard the PNSY sponsor will submit the NAVSEA LOCAL VISIT REQUEST via the PNSY homepage.
- 6.4.2. If employees do not have a government issued contractor CAC which will allow access to the base, a SECNAV 5512/1 will need to be filled out and brought in with them to Pass & ID in order for the individual to be vetted and granted a (DBIDS) Defense Biometric Identification System pass or credential to allow access to the base.
- 6.4.3. Personnel must provide two forms of ID. One of which must prove U.S. citizenship in order to access the NAVSEA Controlled Industrial Area (CIA) or any other NAVSEA spaces. Documents: a valid REAL ID compliant driver's license, passport, birth certificate or government CAC. All documents must be active. No expired documents will be accepted.
- 6.4.4. Badge Criteria:
 - 6.4.4.1. Green badges will be issued to contractors who are verified U.S. citizens, have a shipyard sponsor and a verified need to provide a service or perform work at PNSY.
 - 6.4.4.2. These badges provide unescorted access to the CIA, excluding Nuclear Work Area's (NWA's), Controlled Nuclear Information Areas (CNIA's), Security Islands (SI's) and Other Sensitive Areas (OSA)
 - 6.4.4.3. The Standard Access Control Badge (SACB) is U.S. Government property and it must be protected and returned to Pass & ID office or Security Department upon termination of employment, completion of work, service performed, expiration of contract or no longer a need to enter NAVSEA controlled spaces.
 - 6.4.4.4. The SACB will be displayed at the chest level area of the outer clothing with the picture side visible at all times.
 - 6.4.4.5. No objects, pins or stickers can be attached to the badge. If the badge becomes faded, damaged or illegible for any reason the bearer will report to the Pass & ID office for a new badge.
- 6.4.5. Camera-enabled devices of any kind, including cell phones, laptops, and tablets, are prohibited inside the CIA. Laptops without cameras must be registered with the Security Office before entering the CIA.
- 6.4.6. All contractor vehicles are subject to inspection prior to allowing entry onto this installation.
- 6.4.7. Contractor vehicles may enter the CIA through the open CIA vehicle gate. Personal vehicles are precluded from entry. CIA access by Contractor vehicles will be kept to the absolute minimum necessary to accomplish the contracted work. All contractor vehicles must meet the following:
 - 6.4.7.1. Possess a valid vehicle registration showing the vehicle is a company owned or leased vehicle, or rental vehicle rented by the company; a valid state safety inspection as required by the state of registration and an inspection pass from Commercial Vehicle Inspections Station (CVIS) located at the main gate.
 - 6.4.7.2. Contractor logo should be located on both sides of the vehicle and either painted on or magnetic sign. Paper or cardboard signs are not authorized. Lettering must be a minimum of 2-inch font and displayed in a manner that is clearly visible.

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6.4.7.3. Contractor vehicles will be authorized access to the CIA to deliver contractor equipment to the work site. The use of these vehicles solely for the transport of personnel is not authorized.

6.4.8. CIA inspection passes will be issued to a contractor vehicle meeting the above criteria. CIA passes will have an expiration date, time and initials of the issuing officer. From the time of issue, the contractor vehicle must enter the CIA within 7 minutes from the time issued. Any contractor attempting to access the CIA after the 7-minute time window will be denied entry and directed back to CVIS for re-inspection.

6.5. **Receipt Inspection Criteria** – Equipment must meet all criteria defined in this specification.

6.6. **Return** - The Government will provide a temporary storage location convenient for Contractor pick-up. The Government will notify the Contractor as well as the Contracting Office as to when the rental Item has been released and is available for pick-up. The last day of the rental period shall be the next normal workday after the Contractor has been so notified. The Contractor shall be responsible for shipping. The Government can provide forklift services during work hours for placing the equipment onto the contractor's transport.

7. **GENERAL**

7.1. **Response to Request** - As part of the response to this request, descriptive literature shall be furnished in sufficient detail to show that the proposed design will meet these specifications. Contractor submittals shall include brochures of the model being submitted, assembly sketches with critical dimensions, sketches (with dimensions) of all tooling being provided, statements of compliance with specification, and performance statements with special instructions. Supporting information shall be organized and packaged by unit. If supporting information includes multiple models the Contractor shall clearly state model number of proposed unit and redact all information not applicable to proposed unit.

8. **ATTACHMENTS**

8.1. N/A

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Enclosure 1

Table E1

SOW Compliance Checklist

SOW	REQUIREMENT	COMPLIANCE WITH SOW. (See Instructions Below)	EXCEPTIONS TO SOW. (See Instructions Below)
Equipment			
Table 1	Equipment Specification (HVAC AHU)		
Table 2	Equipment Specification (Exh Blower)		
Table 3	Equipment Specification (DCCU)		
Table 4	Equipment Specification (Elec Heater)		
Table 5	Emergency Remote Stop for Air Equip.		
Table 6	Equipment Specification (Chiller)		
Table 7	Equipment Specification (Pump)		
Table 8	Equipment Specification (Hoses)		
Section 4.1-4.2	Chiller System Capabilities		
Section 4.3	Hydrostatic Testing		
Section 4.4	Phase Rotation Indication		
Section 4.5-4.6	Accessories		
Section 4.7	HVAC System Capabilities		
Section 4.8	DCCU System Capabilities		
Section 4.9	Construction & Weather Hardiness		
Section 4.10	Electrical (NRTL, NEMA)		
Section 4.11	Lifting & Handling		
Section 4.12	Noise		
Section 4.13-4.14	Tagging/Marking		
Section 4.15	Electrical Safety		
SOW	REQUIREMENT	COMPLIANCE WITH SOW. (See Instructions Below)	EXCEPTIONS TO SOW. (See Instructions Below)
Services			
Section 5 (in its entirety)	Deliverables		
Section 6 (in its entirety)	Delivery Acceptance and Return		
INSTRUCTIONS			
1. COMPLIANCE WITH SOW. (i) Use YES or NO to identify if equipment and services comply with all SOW requirements. (ii) Ensure answers are consistent with any brochures, specification sheets or other descriptive literature provided in your quote.			
2. EXCEPTIONS TO SOW. (i) Identify any exceptions to the SOW. Note, answers are required for any NO in prior column. (ii) Explain reason for exception.			